

Few-Shot Open-Set Keyword Spotting at Vocabulary Scale:

A Metric-Learning Study of Encoders, Losses, and Open-Set Rejection

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Abstract. We study *few-shot open-set keyword spotting* (KWS): a single embedding encoder is trained once on a large word vocabulary and is then deployed to recognize an *arbitrary* user-defined keyword set from only a handful of enrollment examples, while rejecting out-of-vocabulary speech and silence. We build a fully reproducible pipeline around two compact encoders — a depthwise-separable CNN (DSCNN-L, ≈ 0.41 M parameters) and a temporal-attention edge model (EdgeSpot-Full $\tau=4$, ≈ 0.13 M parameters) — combined with three feature front-ends (MFCC, log-mel, and a trainable per-channel PCEN) and four metric-learning objectives (Triplet, Sub-center ArcFace (SCAF), Generalized End-to-End (GE2E), and a cosine knowledge-distillation (KD) objective from a frozen Wav2Vec 2.0 teacher). All systems are trained on Multilingual Spoken Words Corpus (MSWC) English and evaluated cross-corpus on Google Speech Commands v2 (GSC) under a strict *EdgeSpot-exact* 10-shot open-set protocol with 100 randomized runs. Through a 16-configuration factorial screen we show that (i) the trainable PCEN front-end is the single most reliable improvement (up to +9.2 accuracy points at 1% false-accept rate for the edge encoder); (ii) the centroid-based GE2E objective dominates classification-style losses for open-set generalization; and (iii) crucially, the classification objective SCAF *collapses* when the training vocabulary grows to 37,387 classes unless its loss weight is reduced by four orders of magnitude. We further characterize how accuracy *saturates* near two million training clips and show that the cosine-KD objective reaches GE2E-level accuracy with roughly half the data. The best overall configuration (DSCNN-L + PCEN + GE2E) attains 86.36% open-set accuracy at 1% false-accept rate and 95.21 AUC on GSC test, while the best compact model (EdgeSpot-Full + PCEN + GE2E) reaches 82.87% at one third of the parameters. We release the complete code, training/evaluation protocols, and an offline web demonstrator.

Keywords: keyword spotting, few-shot learning, open-set recognition, metric learning, prototypical networks, GE2E, PCEN, knowledge distillation, edge AI.

1 Introduction

Keyword spotting (KWS) — the task of detecting a small set of spoken commands in a continuous audio stream — is the always-on entry point of nearly every modern voice interface. Classical KWS systems are trained as *closed-set* classifiers over a *fixed* vocabulary (e.g., the ten Google Speech Commands words). This design has two structural limitations. First, the keyword set is frozen at training time: supporting a new wake word or command requires collecting data and retraining the model. Second, a softmax classifier is, by construction, *not* open-set: every input is forced into one of the known classes, so unseen words and background noise are silently misclassified as commands, producing false activations.

This work targets the more realistic and more difficult **few-shot open-set** formulation. We train a single *embedding encoder* on a very large and diverse word vocabulary, and at deployment time the end user defines an *arbitrary* keyword set by providing only a few (k) enrollment utterances per word. Each keyword is represented by a *prototype* (the mean embedding of its enrollment shots); recognition is performed by nearest-prototype matching in embedding space, and an explicit open-set decision rejects anything that is too

far from all prototypes. Because the encoder is never retrained for a particular keyword, the same model serves unlimited downstream vocabularies; because rejection is distance-based, the system can abstain on out-of-vocabulary speech and silence.

The central scientific question we pursue is *what makes such an encoder generalize*, and in particular how the design choices interact with the *scale* of the training vocabulary. We perform a controlled factorial study over three axes — feature front-end, encoder backbone, and metric-learning objective — and we deliberately push the training vocabulary from a 31-word “micro-set” up to the full 37,387-word MSWC English corpus. This scaling exposes a phenomenon that is invisible at small scale: a classification-style angular-margin loss that is competitive on a few hundred classes *catastrophically collapses* on tens of thousands of classes.

Contributions. Our main contributions are:

1. A reproducible, dependency-light few-shot open-set KWS framework with two compact encoders, three feature front-ends, four metric objectives, and four open-set rejection heads (nearest-class-mean / Open-Max / energy / calibrated), all sharing a common prototype interface (Sections 4 and 5).

2. A 16-configuration factorial screen on full-vocabulary MSWC that isolates the contribution of each axis with paired deltas; PCEN and GE2E emerge as the two robust wins (Section 8.1).
3. A scale analysis showing (a) accuracy saturation near ~ 2 M training clips and (b) the failure mode of Sub-center ArcFace at 37k classes, with a gradient-magnitude explanation and a practical fix (Section 8.2).
4. A knowledge-distillation study showing that a cosine objective against a frozen Wav2Vec 2.0 teacher matches GE2E accuracy with roughly half the training data (Section 8.3).
5. A complete cross-corpus evaluation under a strict EdgeSpot-exact 10-shot protocol (100 runs), together with a CPU streaming pipeline (Silerio VAD + smoothing state machine) and an offline web demonstrator (Section 9).

2 Related Work

Small-footprint KWS. Depthwise-separable convolutions popularized by DS-CNN [1] remain a strong accuracy/size trade-off baseline for on-device KWS, later refined by broadcasted-residual networks (BC-ResNet) [2]. Our DSCNN-L encoder follows the DS-CNN size-table design, and our EdgeSpot encoders combine fused/broadcasted temporal blocks with a single-head temporal self-attention stage in the spirit of attention-augmented edge KWS.

Few-shot and metric-learning KWS. Treating KWS as metric learning enables flexible vocabularies. Prototypical networks [3] and triplet embeddings [4] provide the classic few-shot machinery; GE2E [5] was introduced for speaker verification and naturally mirrors the enroll-then-classify deployment of few-shot KWS. We adopt GE2E as a *centroid* objective and compare it head-to-head with triplet and classification objectives.

Angular-margin classification. ArcFace [6] and its sub-center variant [7] produce highly discriminative embeddings for face recognition with thousands of identities. We test whether the same machinery transfers to word embeddings at 10^4 -class scale and report a negative result with a concrete remedy.

Open-set recognition. OpenMax [8] rejects unknowns by Weibull tail modeling of activations, and energy scores [9] provide a calibration-free alternative. We adapt both to operate on prototype distances and compare them to a simple nearest-class-mean (NCM) distance threshold.

Front-ends and self-supervision. PCEN [10] is a trainable, per-channel adaptive gain/compression front-end designed for robust far-field detection. Wav2Vec 2.0 [11] provides powerful self-supervised

speech representations; we use a frozen Wav2Vec 2.0 as a distillation teacher rather than as the deployed model, keeping the student tiny.

3 Problem Formulation

Let $f_\theta : \mathbb{R}^{T \times F} \rightarrow \mathbb{R}^d$ be an encoder mapping a one-second log-frequency representation to a unit-norm embedding, $f_\theta(x) = \tilde{z}/\|\tilde{z}\|_2$. At deployment the user supplies a *support set* of k utterances per keyword $w \in \mathcal{K}$. The prototype of w is the mean support embedding

$$p_w = \frac{1}{k} \sum_{i=1}^k f_\theta(x_i^{(w)}). \quad (1)$$

For a query x we compute the nearest prototype and its distance,

$$\hat{w}(x) = \arg \min_{w \in \mathcal{K}} d(f_\theta(x), p_w), \quad s(x) = -\min_w d(f_\theta(x), p_w), \quad (2)$$

with d the Euclidean distance. The *open-set decision* accepts x as the keyword $\hat{w}(x)$ iff the score exceeds an operating threshold τ , and otherwise emits **unknown**:

$$\text{decision}(x) = \begin{cases} \hat{w}(x), & s(x) \geq \tau, \\ \text{unknown}, & s(x) < \tau. \end{cases} \quad (3)$$

The encoder θ is trained on a *disjoint* corpus and vocabulary; at test time $\mathcal{K}$ and the threshold τ are chosen *without* updating θ . The two coupled goals are (i) high keyword accuracy on the known set and (ii) high rejection of unknown words and silence — exactly the detection/false-alarm trade-off captured by the DET curve.

4 System Architecture

Figure 1 summarizes the end-to-end pipeline. We now describe each stage with the exact configuration used in our experiments.

4.1 Feature front-ends

All audio is mono, 16 kHz, and trimmed/zero-padded to exactly one second (16,000 samples). We support two spectral parameterizations, summarized in Table 1.

MFCC (DSCNN front-end). We compute 40 Mel-frequency cepstral coefficients with a 1024-point FFT, 640-sample (40 ms) Hann window, 320-sample (20 ms) hop, and `center=False`; the Mel filterbank uses Slaney scale and Slaney area normalization, and the log is taken as $10 \log_{10}(\cdot)$ followed by an orthonormal DCT-II. We retain *only the first 10 coefficients* and transpose to a tensor of shape $(1, 47, 10) = (\text{channel}, T_{\text{frames}}, \text{features})$, where $T = 47 = \lfloor (16000 - 1024)/320 \rfloor + 1$.

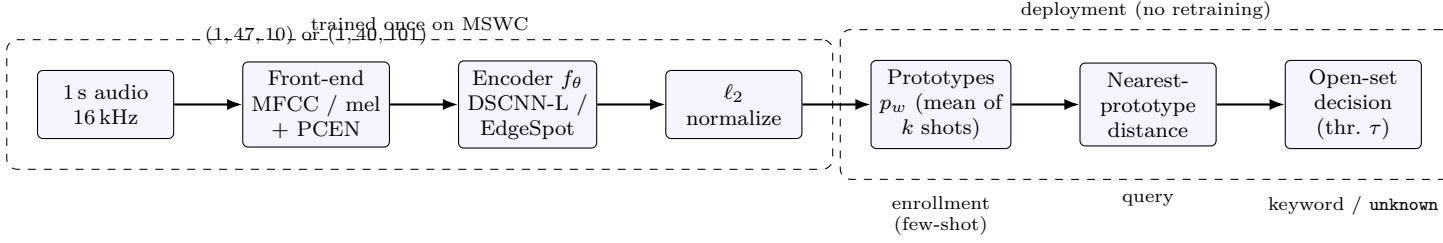


Figure 1: End-to-end few-shot open-set KWS pipeline. The encoder is trained once on MSWC; at deployment, user keywords are enrolled as prototypes and queries are classified by nearest-prototype distance with an explicit open-set threshold.

Table 1: Feature front-end configurations. Both produce a one-channel time $\times$ feature map; PCEN is an optional trainable layer applied after the spectral stage.

	MFCC	Log-mel
Sample rate	16 kHz, mono, 1 s	
FFT size	1024	512
Window / hop	40/20 ms	25/10 ms
center	False	True
Mel bands	40	40
Output coeffs.	10 (first DCT)	40 (mel)
Tensor shape	(1, 47, 10)	(1, 40, 101)
Trainable PCEN	optional	optional

Log-mel (EdgeSpot/BC-ResNet front-end). We use 40 Mel bands with a 512-point FFT, 400-sample (25 ms) window, 160-sample (10 ms) hop and **center=True**, producing (1, 40, 101). The log compression is delegated to the PCEN layer (below).

Trainable PCEN. Per-channel energy normalization [10] applies a causal first-order IIR smoother $M_t = (1 - s)M_{t-1} + sE_t$ followed by adaptive gain and dynamic-range compression,

$$\text{PCEN}(E_t) = \left(\frac{E_t}{(\epsilon + M_t)^\alpha} + \delta \right)^r - \delta^r, \quad (4)$$

with learnable α, δ, r, s (initialized to 0.96, 2.0, 0.5, 0.04; $\epsilon=10^{-6}$). PCEN replaces the static log and adapts to channel statistics during training.

4.2 Encoders

DSCNN-L. A depthwise-separable CNN built from a size table: an initial Conv(1 $\rightarrow$ 276, 10 $\times$ 4, stride 2 $\times$ 1) with TensorFlow-style asymmetric “same” padding, followed by five depthwise separable blocks (276 channels, 3 $\times$ 3 kernels; the first block strides 2 $\times$ 2). All but the last block use BatchNorm+ReLU; the last block uses an (affine-free) LayerNorm. A global average pool yields the embedding directly, so the embedding dimension equals the channel width, $d = 276$ (there is no final linear projection). DSCNN-L has ≈ 0.413 M parameters.

EdgeSpot-Full ($\tau=4$). A width-scaled edge encoder operating on the (1, 40, 101) mel/PCEN input. A 5 $\times$ 5 stride-(2, 1) stem feeds four stages of fused-temporal and

Table 2: Encoder summary. Embedding dimension d , parameter count, and front-end. Parameter counts are measured from the instantiated models.

Encoder	Front-end	d	Params
DSCNN-L	MFCC / mel(+PCEN)	276	0.413 M
EdgeSpot-Full $\tau=4$	mel(+PCEN)	64	0.131 M
EdgeSpot-Lite	mel(+PCEN)	64	< 0.131 M
BC-ResNet-FS $\tau=4$	mel(+PCEN)	64	~ 0.1 M

broadcasted-residual “lite” blocks with increasing temporal dilation (1, 2, 4, 8); a depthwise frequency-collapse block averages over frequency to a (C, T) sequence; a depthwise positional convolution and a single-head temporal self-attention block ($d = 64$) model long-range temporal structure; finally a depthwise temporal head and global pooling produce a 64-dimensional embedding. EdgeSpot-Full $\tau=4$ has ≈ 0.131 M parameters, $\sim 3.2\times$ smaller than DSCNN-L. A lighter *EdgeSpot-Lite* variant and a clean *BC-ResNet-FS* baseline (no attention) are also provided.

Table 2 compares the encoders. ℓ_2 -normalization is applied *outside* the encoder so that all losses and the deployment classifier operate on the hypersphere.

4.3 Metric-learning objectives

All objectives are trained *episodically*: each episode samples $n_{\text{cls}}=30$ classes with $n_{\text{spp}}=20$ utterances each (600 embeddings/episode).

Triplet. Semi-hard triplet loss [4] with Euclidean distance, $\mathcal{L} = \frac{1}{N} \sum \max(0, d(a, p) - d(a, n) + m)$, hardest-positive mining and FaceNet semi-hard negative mining.

Sub-center ArcFace (SCAF). An angular-margin softmax with $K=3$ sub-centers per class. With normalized weights $W \in \mathbb{R}^{(C \cdot K) \times d}$, the per-class logit is the *max* cosine over its sub-centers, $\cos \theta_c = \max_{j \leq K} w_{c,j}^\top z$; the additive margin $m=0.5$ is applied to the target class and logits are scaled by $s=30$ before cross-entropy.

GE2E (centroid). Generalized end-to-end loss [5] that mirrors deployment: within an episode each class is split into support/query; the centroid is the ℓ_2 -normalized support mean and queries are classified

by scaled cosine similarity to all centroids with learnable scale and bias, optimized by cross-entropy. GE2E directly optimizes the enroll-then-match objective of Equations (1) and (2).

Cosine knowledge distillation (KD). We additionally distill from a *frozen* Wav2Vec 2.0-base teacher (hidden layer 16, mean-pooled, projected to $d=64$ by a sub-center-ArcFace-trained head). The distillation loss is a pure embedding-cosine term,

$$\mathcal{L}_{\text{KD}} = 1 - \frac{1}{N} \sum_i \cos(f_\theta(x_i), g_{\text{teacher}}(x_i)), \quad (5)$$

i.e. *no* temperature/soft-label term. Teacher embeddings are precomputed offline and looked up by file path during training.

Hybrid combinations. We form weighted sums, e.g. `scaf_ge2e`, `kd_ge2e`, `kd_scaf`, `kd_scaf_ge2e`. A crucial implementation detail (and a key empirical finding, Section 8.2) is the SCAF weight: under KD it is set to 5×10^{-5} while GE2E/KD weights are 1.0. The very small SCAF weight prevents the high-magnitude angular-margin classification loss from dominating the centroid/cosine terms.

5 Open-Set Rejection

All rejection heads share the same prototype interface and differ only in the *rejection score*; the predicted keyword is always the argmin-distance class (Equation (2)).

Nearest-class-mean (NCM, “Direct ℓ_2 ”). Reject iff the minimum prototype distance exceeds a threshold. This is our default and reflects the thesis’s “direct distance” contribution: it uses *raw* distances with no probability normalization, in contrast to a softmax over negative distances (the probability baseline).

OpenMax. Per class, fit a Weibull distribution to the tail of support-to-prototype distances; the membership score is $1 - \text{CDF}_{\text{Weibull}}(d(x, p_c))$, optionally blended with $\exp(-d)$ [8].

Energy OOD. Treat $-d_i$ as logits and use the free energy $E(x) = T \cdot \log \sum_i \exp(-d_i/T)$ as the acceptance score [9]; the single temperature T interpolates between log-sum-exp and the ℓ_2 baseline.

Calibrated thresholds. For analysis we also compute support-uncertainty thresholds $\mu + \alpha\sigma$ ($\alpha=2$) of support distances and impostor-FAR thresholds; these drive per-word confusion tables but not the headline threshold-free metrics.

6 Datasets and Training Protocol

Training corpus (MSWC English). We train on the English subset of the Multilingual Spoken Words Corpus [12], which after filtering for $\geq$ minimum clips contains 37,387 training words (and 763 validation words), drawn from a metadata pool of $\sim 38,150$ words and ~ 6.6 M clips. To study scale we form caps of $\{20, 50, 220, 620\}$ clips per word — roughly $\{0.53, 0.94, 2.05, 2.99\}$ M training clips — and two reduced regimes: a *Top-500* legacy split (450 train / 50 val words, evaluation on words ranked 501+ with ≥ 1000 utterances) and an MLCommons *micro-set* (31 keywords) used only for pipeline validation and architecture screening.

Episodic sampling and augmentation. Episodes draw 30 classes $\times$ 20 utterances (only classes with ≥ 20 clips are eligible). Online augmentation mixes DE-MAND background noise (probability 0.5, SNR sampled in $[0, 10]$ dB) and applies SpecAugment (one frequency mask of width 6, one time mask of width 8); gain perturbation (± 6 dB) and time-shift (≤ 100 ms) are applied with probability 0.5.¹

Evaluation corpus (GSC v2). We evaluate *cross-corpus* on Google Speech Commands v2 [13]. Enrollment (support) samples are drawn from the official `validation_list.txt`; query samples from `testing_list.txt` (“test”) or the remaining files (“dev”), preventing leakage. A synthetic `_silence_` class is generated from random one-second crops of `_background_noise_`, with disjoint crops for support and query.

EdgeSpot-exact protocol. The headline protocol partitions GSC’s 35 words into 11 positive classes — the ten command words $\{yes, no, up, down, left, right, on, off, stop, go\}$ plus `_silence_` — and 25 negative (unknown) spoken words. Each run enrolls $k=10$ shots per positive word, computes prototypes (Equation (1)), and scores up to 50 query samples per word (a fixed-seed subsample). We report the mean $\pm$ std over 100 randomized runs; randomness comes from enrollment-shot selection (the known/unknown vocabulary itself is fixed across runs). “test100” denotes 100 runs on the test split, “dev30” denotes 30 runs on the dev split.

7 Evaluation Metrics

We treat the open-set problem as a binary keyword-vs-non-keyword detection augmented by a multiclass identity track, and build all metrics on the ROC.

¹The executed configuration uses Adam with $\text{lr}=10^{-3}$, weight decay 10^{-4} , gradient clipping at 5.0, a cosine-annealing-with-warm-restarts schedule ($T_0=10, T_{\text{mult}}=2, \eta_{\text{min}}=10^{-5}$), and seed 42 with cuDNN determinism. Per-experiment epoch/episode budgets are noted in the result tables.

- **AUC**: area under the ROC of the open-set scores (Equation (2)); threshold-free.
- **EER**: equal-error rate, the operating point minimizing $|FAR - FRR|$, reported as $(FAR + FRR)/2$.
- **FRR@FAR**: false-reject rate at the largest threshold whose false-accept rate does not exceed the target (1% or 5%).
- **Open-set ACC@FAR**: the headline accuracy. A known query is correct iff it is *accepted* (score $\geq \tau_{FAR}$) and the argmin label is correct; an unknown query is correct iff it is *rejected*. Reported at $FAR \in \{1\%, 5\%\}$.
- **Keyword ACC**: closed-set nearest-prototype accuracy over known queries only (ignores rejection).
- **F1**: binary keyword-vs-unknown F1 at the EER threshold.

Interpretation caveat. AUC and EER are threshold-free, whereas ACC@FAR and FRR@FAR use a threshold derived from the evaluation ROC (an oracle operating point), not transferred from a held-out calibration split. We therefore treat ACC@FAR as a comparison metric across systems under matched conditions, and report AUC/EER/F1 jointly so that degenerate operating points are visible. A system that rejects nearly everything can score a high open-set ACC against many negatives while missing all positives; such cases are exposed by $FRR \rightarrow 100\%$ and $F1 \rightarrow 0$.

8 Experiments

We organize the empirical study around four questions: (Q1) which design axes matter (Section 8.1); (Q2) how performance scales with vocabulary/data and why a classification loss collapses (Section 8.2); (Q3) can distillation substitute for data (Section 8.3); and (Q4) what is the best deployable system (Section 8.4).

8.1 Q1: Factorial screen of front-end $\times$ backbone $\times$ loss

We train all $2 \times 2 \times 4 = 16$ combinations of $\{\text{DSCNN-L, EdgeSpot-Full}\} \times \{\text{MFCC, PCEN}\} \times \{\text{Triplet, SCAF, GE2E, SCAF+GE2E}\}$ on full-vocabulary MSWC (cap-620) and evaluate on GSC test100 at 1% FAR. Table 3 reports the full matrix.

Paired effect deltas. Holding the other two axes fixed, Table 4 isolates each axis. Two findings are robust: *PCEN consistently improves every backbone* (largest for the edge encoder, +9.22 ACC and +21.86 F1), and *GE2E consistently beats triplet*. The classification objective SCAF, and the SCAF+GE2E hybrid at full SCAF weight, are not merely worse — they *collapse* at this vocabulary scale (Section 8.2).

Table 3: Full 16-configuration screen on full-vocabulary MSWC (cap-620), GSC test100 @ 1% FAR. “coll.” marks degenerate collapse (AUC=50, F1=0, FRR=100%); see Section 8.2. Best in bold.

Enc.	Feat.	Loss	ACC _{1%}	AUC	EER	F1
DSCNN	MFCC	Triplet	71.52	75.55	31.41	57.17
DSCNN	MFCC	SCAF	70.08	55.04	46.41	41.38
DSCNN	MFCC	GE2E	77.08	86.46	21.95	68.50
DSCNN	MFCC	SCAF+GE2E	69.04	47.78	52.15	35.93
DSCNN	PCEN	Triplet	79.98	90.65	17.57	74.14
DSCNN	PCEN	SCAF	69.44	50.00	50.00	coll.
DSCNN	PCEN	GE2E	82.34	92.42	14.89	77.75
DSCNN	PCEN	SCAF+GE2E	69.44	50.00	50.00	coll.
Edge	MFCC	Triplet	69.63	52.84	48.05	39.79
Edge	MFCC	SCAF	69.44	50.00	50.00	coll.
Edge	MFCC	GE2E	70.76	65.30	39.05	48.82
Edge	MFCC	SCAF+GE2E	69.67	50.88	50.39	37.57
Edge	PCEN	Triplet	79.58	89.85	18.22	73.29
Edge	PCEN	SCAF	69.44	50.00	50.00	coll.
Edge	PCEN	GE2E	79.98	87.23	20.23	70.68
Edge	PCEN	SCAF+GE2E	69.44	50.00	50.00	coll.

Table 4: Paired effect deltas (percentage points) on full-vocabulary MSWC, GSC test100 @ 1% FAR. Positive favors the first option.

Contrast (fixed others)	$\Delta\text{ACC}_{1\%}$	ΔAUC	ΔF1
PCEN – MFCC (DSCNN, GE2E)	+5.26	+5.96	+9.25
PCEN – MFCC (Edge, GE2E)	+9.22	+21.93	+21.86
GE2E – Triplet (DSCNN, PCEN)	+2.36	+1.77	+3.61
DSCNN – Edge (PCEN, GE2E)	+2.36	+5.19	+7.07

8.2 Q2: Vocabulary/data scale and the SCAF collapse

Accuracy saturates near 2M clips. Fixing the best front-end/loss (PCEN+GE2E) and sweeping the per-word cap, Figure 2 and Table 5 show a large jump between cap-50 and cap-220 (+3.55 ACC for DSCNN, +3.79 for EdgeSpot at 5% FAR) and a flat region from cap-220 to cap-620 (+0.33 / −0.02). Accuracy is effectively *data-saturated* at ~ 2 M training clips; beyond this point, extra clips of the same words yield negligible cross-corpus gains.

The SCAF collapse at 37k classes. The most striking result of the scale study is a *qualitative* failure. On the 31-class micro-set and the ~ 500 -class Top-500 split, SCAF and SCAF+GE2E are excellent (AUC ≈ 95 ; see Section 8.4). However, on the full 37,387-class vocabulary with the default SCAF weight of 1.0, training *collapses from epoch 2*: validation AUC pins at 0.5000, the GE2E in-episode accuracy falls to ≈ 0.033 ($\approx 1/30$, i.e. chance), and the resulting GSC metrics are degenerate (AUC 50, F1 0, FRR=100%, keyword ACC 9.09%). The mechanism is a gradient-magnitude imbalance: the sub-center ArcFace logit scale ($s=30$) multiplied across $\sim 3.7 \times 10^4$ classes produces a classification gradient that dwarfs the centroid/cosine terms, driving all embeddings to a trivial solution. The same hybrid does *not* collapse under KD because there the SCAF weight is 5×10^{-5} . The practical remedies we verify are: (i) reduce the SCAF weight to $\sim 5 \times 10^{-5}$, (ii) warm up the

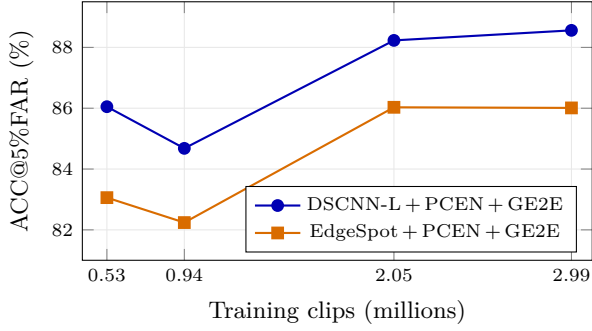


Figure 2: Cross-corpus accuracy on GSC test100 versus training-data volume (per-word caps 20/50/220/620). Performance saturates near ~ 2 M clips.

Table 5: Data-scale sweep (PCEN+GE2E), GSC test100. ACC at 5% FAR.

Encoder	Cap	ACC _{5%}	AUC	EER	F1
DSCNN-L	20	86.05	91.57	16.25	75.90
	50	84.68	90.45	17.42	74.34
	220	88.23	93.87	12.78	80.67
	620	88.56	94.04	12.53	81.02
EdgeSpot	20	83.06	87.22	20.40	70.46
	50	82.24	87.74	20.19	70.73
	220	86.03	91.31	16.47	75.61
	620	86.01	91.34	16.64	75.38

margin/scale, or (iii) drop the classification term entirely and use GE2E or KD. This is a cautionary result: *angular-margin classification losses that are state-of-the-art at 10^2 – 10^3 classes do not transfer naively to 10^4 -class word embedding.*

8.3 Q3: Knowledge distillation trades teacher knowledge for data

We distill a frozen Wav2Vec 2.0 teacher into the compact EdgeSpot student via the cosine objective of Equation (5) (loss `kd_scaf`). Table 6 shows that at cap-50 KD improves every metric over GE2E (+3.60 ACC@1%FAR, -4.78 EER, +6.31 F1) and, strikingly, *KD at cap-50 matches GE2E at cap-220* — the same accuracy with roughly half the training clips. KD also saturates early (cap-50 $\rightarrow$ cap-220 is essentially flat), indicating that the teacher injects most of its useful inductive bias from limited data.

8.4 Q4: Best deployable systems

Table 7 consolidates the strongest configurations per data regime. Three operating points are worth highlighting:

- **Best overall accuracy:** DSCNN-L + PCEN + GE2E (cap-620, composite epoch selection) reaches 86.36 ± 1.29 open-set ACC@1%FAR and 89.93 ± 0.65 at 5% FAR, with AUC 95.21, EER 11.32, F1 82.73, keyword ACC 92.92%.
- **Best compact:** EdgeSpot-Full + PCEN + GE2E (cap-620) reaches 82.87 ± 1.22 ACC@1%FAR at

Table 6: Knowledge distillation vs. GE2E (EdgeSpot student), GSC test100. KD at cap-50 matches GE2E at cap-220.

Cap	Loss	ACC _{1%}	ACC _{5%}	EER	F1
50	GE2E	77.14	82.24	20.19	70.73
50	KD	80.74	85.82	15.41	77.04
220	GE2E	—	86.03	16.47	75.61
220	KD	80.82	85.90	15.36	77.10

0.131M parameters — within ~ 3.5 points of the best system at one third the size, and competitive with the EdgeSpot-4 literature reference ($\sim 82\%$ ACC@1%FAR, $\sim 128k$ params).

- **Best small-vocabulary:** on the reduced Top-500 split, the EdgeSpot + PCEN + SCAF+GE2E hybrid is the strongest compact model (85.62 ± 1.04 ACC@1%FAR, AUC 95.34), confirming that the hybrid is excellent *when the class count is small enough not to trigger the collapse.*

Reproducibility note on data regimes. Within the full-vocabulary results, three cap-620 runs use different episode budgets (a 40-epoch fixed screen, a long 1000-episode saturation run, and a 300-episode composite-selection development run). They must not be conflated; Tables 3, 5 and 7 draw each row from a single consistent run, and the table captions state the budget where it matters.

9 System Implementation

Beyond the core encoder, the project ships a complete, dependency-light deployment stack. Audio I/O uses `soundfile/scipy` (no torchaudio runtime dependency) to remain compatible with a CUDA-10.2 server.

Denoising (EXT-1). A unified `Denoiser` exposes two backends: a CPU-friendly spectral-gating denoiser (`noisereduce`, with an RMS-guard fallback that prevents over-suppression of short keyword clips) and an optional neural enhancer (SpeechBrain MetricGAN+). Denoising is a runtime toggle and never blocks the core pipeline.

Streaming and VAD (EXT-2). A CPU streaming engine performs sliding-window detection (1s window, 0.5s stride) gated by Silero VAD: only windows that VAD marks as speech are embedded and matched, reducing computation and false activations on silence. A backend-agnostic *state machine* (Algorithm 1) stabilizes raw candidates into clean events using majority-vote smoothing over a short window, a top-2 margin check, and a cooldown to suppress duplicate firings.

Optional speaker gate. An ECAPA-TDNN speaker-verification gate (enroll-then-verify by cosine similarity) can restrict detections to an enrolled speaker; like denoising it is optional and fail-open.

Table 7: Consolidated best systems across data regimes (GSC **test100**, EdgeSpot-exact 10-shot, 100 runs). ACC and FRR at the stated FAR; lower is better for EER/FRR, higher for the rest. “coll.” = degenerate collapse.

Regime (#words)	Encoder	Loss	Params	ACC _{1%}	ACC _{5%}	AUC	EER	F1	KW-ACC
Micro-set (31)	EdgeSpot	SCAF+GE2E (locked)	0.131 M	84.61	86.12	95.61	11.54	82.41	77.66
Top-500 (500)	EdgeSpot	SCAF+GE2E (ep13)	0.131 M	85.62	88.79	95.34	11.51	82.45	90.44
Top-500 (500)	DSCNN-L	GE2E	0.413 M	81.55	86.56	93.17	14.00	78.97	88.62
Full cap-20 (37k)	DSCNN-L	GE2E	0.413 M	82.10	86.05	91.57	16.25	75.90	88.90
Full cap-220 (37k)	DSCNN-L	GE2E	0.413 M	–	88.23	93.87	12.78	80.67	89.82
Full cap-620 (37k)	DSCNN-L	GE2E (ep300)	0.413 M	86.36	89.93	95.21	11.32	82.73	92.92
Full cap-620 (37k)	EdgeSpot	GE2E (ep300)	0.131 M	82.87	86.76	92.41	14.82	77.85	87.29
Full cap-50 (37k)	EdgeSpot	KD	0.131 M	80.74	85.82	91.19	15.41	77.04	87.95
Full cap-220 (37k)	EdgeSpot	SCAF+GE2E ($w=1$)	0.131 M	coll.	coll.	50.00	50.00	coll.	9.09

Algorithm 1 Streaming decision state machine (smoothing + cooldown).

Require: window W , votes V , margin $m_{\min}$, cooldown C

```

1: buffer  $\leftarrow$  deque(maxlen =  $W$ );  $t_{\text{last}} \leftarrow -\infty$ 
2: for each candidate  $c$  with  $[t_s, t_e]$  do
3:   if  $t_s - t_{\text{last}} < C$  then
4:     emit COOLDOWN; continue
5:   end if
6:   if  $\neg c.\text{detected}$  or  $c.\text{kw}=\text{unknown}$  or  $c.\text{margin} < m_{\min}$  then
7:     clear buffer; emit REJECTED; continue
8:   end if
9:   push  $c$ ;  $k^* \leftarrow$  majority keyword in buffer
10:  if  $\text{votes}(k^*) \geq V$  then
11:     $a \leftarrow \arg \max_{\text{conf}} \{c \in \text{buffer} : c.\text{kw}=k^*\}$ 
12:     $t_{\text{last}} \leftarrow a.t_e$ ; clear buffer; emit DETECTED( $k^*$ )
13:  else
14:    emit SCORING
15:  end if
16: end for

```

Web demonstrator. A FastAPI backend and a React front-end provide an offline demo with hot-swappable checkpoints (auto-discovery, path loading, and direct file upload), long-audio timing analysis, and a sampled 17-known/17-unknown open-set evaluation. We stress that the in-UI sampled evaluation is a calibration/illustration tool and does *not* replace the EdgeSpot-exact test-set protocol used for all reported numbers.

Latency budget. The streaming pipeline is designed to a per-component CPU budget (VAD < 5 ms/chunk, MFCC < 5 ms, encoder < 10 ms, classifier < 1 ms; core < 21 ms; < 51 ms with denoising; < 71 ms with speaker verification; < 100 ms end-to-end). These are design targets; a measured field benchmark (false-alarms/hour, miss rate, wall-clock latency on target hardware) is left to future work.

10 Discussion

Why GE2E and PCEN win. GE2E optimizes the *same* enroll-then-match objective used at deployment (Equations (1) and (2)), so the train/test objective gap is minimal; triplet optimizes a proxy (relative pairwise margins) and classification losses optimize a closed-set decision boundary that is discarded at test time. PCEN’s per-channel adaptive compression normalizes

the large dynamic range of cross-corpus recordings (MSWC→GSC), which matters more for the smaller edge encoder (largest PCEN gain), consistent with the front-end carrying relatively more of the representational burden when the backbone is small.

Why classification collapses at scale. The collapse is not a tuning artifact but a structural mismatch: angular-margin softmax couples *all* classes through a single normalized weight matrix, so its gradient norm grows with class count, while GE2E/cosine-KD gradients are bounded per episode/sample. At 37k classes the classification term overwhelms the metric terms unless down-weighted by $\sim 10^{-4}$ or removed. This generalizes a known practical lesson from face recognition (where careful scale/margin scheduling is essential at large identity counts) to large-vocabulary word embedding.

Capacity vs. data. DSCNN-L ($3.2\times$ larger) consistently leads on accuracy, but the gap to the compact EdgeSpot (~ 3.5 ACC points) is small relative to the size saving, and KD narrows the data requirement of the compact model. For tight edge budgets, EdgeSpot + PCEN + GE2E (optionally KD-initialized) is the recommended operating point; for best accuracy, DSCNN-L + PCEN + GE2E.

11 Limitations

(i) ACC@FAR/FRR@FAR use ROC-derived (oracle) thresholds rather than thresholds transferred from an independent calibration split; we mitigate this by always co-reporting threshold-free AUC/EER/F1. (ii) Enrollment is fixed at $k=10$ shots throughout; a k -shot sensitivity sweep ($k \in \{1, 3, 5\}$) is not included. (iii) The streaming/denoising latency figures are design budgets, not measured field benchmarks, and there is no quantitative denoising ablation (SNR-sweep accuracy). (iv) Some large-vocabulary runs were epoch-limited by cloud-session constraints; the Top-500 epoch-25 point is reported from training logs rather than a locked artifact and is excluded from headline claims.

12 Conclusion

We presented a controlled study of few-shot open-set keyword spotting trained at true vocabulary scale (up to 37,387 words) and evaluated cross-corpus under a strict 10-shot, 100-run protocol. The trainable PCEN front-end and the centroid GE2E objective are the two robust, transferable wins; a frozen-teacher cosine distillation halves the data needed by the compact encoder; and, most importantly, we documented and explained a catastrophic collapse of sub-center ArcFace at 10^4 -class scale, with a simple weighting fix. The best system (DSCNN-L + PCEN + GE2E) attains 86.36% open-set accuracy at 1% FAR and 95.21 AUC on GSC, while a $3.2\times$ smaller edge model stays within a few points. We release code, protocols, and an offline demonstrator to support reproduction. Future work includes a k -shot sensitivity study, a measured on-device latency/false-alarm benchmark, an SNR-sweep denoising ablation, and warm-up/curriculum schedules to stabilize angular-margin losses at scale.

Reproducibility

All experiments use fixed seeds (seed 42, cuDNN determinism). Training, evaluation, and the EdgeSpot-exact protocol are scripted (`scripts/train.py`, `scripts/evaluate.py`, `scripts/evaluate_edgespot_protocol.py`); per-run JSON artifacts store the seed, positive/negative word partition, and all metrics for each of the 100 runs. The web demonstrator runs fully offline.

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